# God's plan MTH 102 practice question

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March 2026

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Functions and Limits: MCQs and Shortcut Solutions

## PART I: FUNCTIONS AND LIMITS

1. A function is said to be an even function of $x$ when $f(-x)$ equals:
   [label=(A)]
   (a) $f(-1)$
   (b) $-f(x)$
   (c) $f(-x)$
   (d) $f(x)$

   **Answer:** (D)
   **Strategy (Definition):** By definition, an even function is symmetrical across the y-axis, meaning a negative input gives the exact same output as a positive input. Therefore, $f(-x) = f(x)$.

2. A function $f(x)$ is said to be an odd function when $f(-x)$ equals:
   [label=(B)]
   (a) $-f(x)$
   (b) $f(-x)$
   (c) $f(x)$
   (d) $f(1)$

   **Answer:** (A)
   **Strategy (Definition):** An odd function has rotational symmetry around the origin. Inputting a negative $x$ pulls the negative sign out to the front: $f(-x) = -f(x)$.

3. Find $f(0)$ in the equation $f(x) = \frac{4x-3}{1+7x}$
   [label=(C)]

   (a) 3

   (b) -2

   (c) -3

   (d) 0

   **Answer:** (C)
   **Strategy (Direct Substitution):** Replace every $x$ with 0.
   $\frac{0-3}{1+0} = -3$

4. If $f(x) = \frac{(x+h)^2-x^2}{h}$, evaluate $\lim_{h\to 0} f(x)$.
   [label=(D)]

   (a) $x + h$

   (b) $2x$

   (c) $x^2$

   (d) $2x + h$

   **Answer:** (B)
   **Strategy (The Derivative Shortcut):** You *could* expand the numerator, but the ultimate shortcut is recognizing that this limit is the literal definition of a derivative! It is asking for the derivative of $x^2$. By the power rule, the derivative of $x^2$ is simply $2x$.

5. Find $\lim_{x\to 5} f(x)$ for $f(x) = \frac{4x-3}{1+7x}$
   [label=(E)]

   (a) 17/34

   (b) 17/14

   (c) 17/36

   (d) 1/3

   **Answer:** (C)
   **Strategy (Direct Substitution):** Always plug the number in first.
   Numerator: $4(5) - 3 = 17$
   Denominator: $1 + 7(5) = 36$
   Result: 17/36. No advanced rules needed.

6. Find $f(1/7)$ in the equation $f(x) = \frac{4x-3}{1+7x}$
   [label=(F)]

   (a) 17/34

   (b) $-17/14$

(c) 17/36

(d) 1/3

**Answer:** (B)
**Strategy (Direct Substitution):**
Numerator: $4(1/7) - 3 = 4/7 - 21/7 = -17/7$
Denominator: $1 + 7(1/7) = 1 + 1 = 2$
Divide the two: $(-17/7)/2 = -17/14$.

7. Evaluate $\lim_{x\to 1} x^2 + 6x + 4$
   [label=(G)]

   (a) 17

   (b) 7

   (c) 17

   (d) 11

   **Answer:** (D)
   **Strategy (Direct Substitution):** This is a continuous polynomial, so just plug in $x = 1$.
   $1^2 + 6(1) + 4 = 11$.

8. Evaluate $\lim_{x\to 3} \frac{x^3+4}{1-x}$
   [label=(H)]

   (a) $-31/2$

   (b) 31/2

   (c) $-68/3$

   (d) 68/3

   **Answer:** (A)
   **Strategy (Direct Substitution):** Plug in $x = 3$.
   Numerator: $3^3 + 4 = 27 + 4 = 31$
   Denominator: $1 - 3 = -2$
   Result: $-31/2$.

9. Evaluate $\lim_{x\to -2/3} \frac{3x^2+5x+2}{3x^2-4x-4}$
   [label=(I)]

   (a) $-31/2$

   (b) 31/2

   (c) $-1/8$

   (d) 1/36

**Answer:** (C)
**Strategy (L'Hôpital's Rule):** Plugging in $-2/3$ gives $0/0$. When you hit $0/0$, immediately use L'Hôpital's Rule (derive top, derive bottom).
Derivative of top: $6x + 5$
Derivative of bottom: $6x - 4$
Plug in $-2/3$: $\frac{6(-2/3)+5}{6(-2/3)-4} = \frac{-4+5}{-4-4} = -1/8$.

10. Evaluate $\lim_{x\to 2}\left(\frac{x}{x-2} - \frac{4}{x^2-2x}\right)$
    [label=(J)]
    (a) 2
    (b) -2
    (c) -1
    (d) 1/2

**Answer:** (A)
**Strategy (Algebraic Cleanup First):** Plugging in 2 gives $\infty - \infty$, which is indeterminate. Combine the fractions first to use L'Hôpital.
Notice $x^2 - 2x = x(x-2)$. Multiply the first fraction by $x/x$ to get common denominators: $\frac{x^2-4}{x(x-2)}$.
Now, plug in $x = 2$ to get $0/0$. Use L'Hôpital!
Derive top: $2x$
Derive bottom (expand to $x^2 - 2x$ first): $2x - 2$
Plug in 2: $2(2)/(2(2) - 2) = 4/2 = 2$.

11. Evaluate $\lim_{x\to 2}\left(\frac{x}{x-2} - \frac{4}{x^2-2x}\right)$
    [label=(K)]
    (a) 2
    (b) -2
    (c) -1
    (d) 1/2

**Answer:** (A)
*(Note: This is an exact duplicate of Question 10 printed on the exam paper)*

12. Evaluate $\lim_{x\to 3}\left(\frac{1}{x-3} - \frac{2}{x^2-4x+3}\right)$
    [label=(L)]
    (a) 2
    (b) -2
    (c) 1

(d) 1/2

**Answer:** (D)
**Strategy (Algebraic Cleanup First):** This yields $\infty - \infty$. Find a common denominator. The right denominator factors to $(x-3)(x-1)$. Multiply the left fraction by $(x-1)/(x-1)$:
$\frac{x-1-2}{(x-3)(x-1)} = \frac{x-3}{(x-3)(x-1)}$.
The $(x-3)$ terms cancel perfectly, leaving $\frac{1}{x-1}$. Plug in $x=3$ to get 1/2.

13. Evaluate $\lim_{x\to\infty}\left[\frac{2x^3}{x^3+1}\right]$
[label=(M)]

(a) 0

(b) -1

(c) 2

(d) $\infty$

**Answer:** (C)
**Strategy (Infinity Degree Rule):** The highest power of $x$ on top is 3. The highest power on the bottom is 3. Since the powers are tied, the limit is simply the ratio of the coefficients attached to those highest powers: $2/1 = 2$.

14. Evaluate $\lim_{x\to 0}\frac{\sqrt{2-x}-\sqrt{2+x}}{x}$
[label=(N)]

(a) 1

(b) 0

(c) $\frac{-1}{\sqrt{2}}$

(d) $\sqrt{2}$

**Answer:** (C)
**Strategy (L'Hôpital's Rule):** Plugging in 0 gives 0/0. Derive the top and bottom.
Bottom derivative is 1.
Top derivative requires the chain rule: $\frac{1}{2\sqrt{2-x}}(-1) - \frac{1}{2\sqrt{2+x}}(1)$.
Plug in $x=0$: $\frac{-1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{-2}{2\sqrt{2}} = \frac{-1}{\sqrt{2}}$.

15. Evaluate $\lim_{x\to 0}\frac{\sqrt{2-x}-\sqrt{2+x}}{x}$
[label=(O)]

(a) 1

(b) 0

(c) $\frac{-1}{\sqrt{2}}$

(d) $\sqrt{2}$

**Answer:** (C)
*(Note: This is an exact duplicate of Question 14 printed on the exam paper)*

16. Evaluate $\lim_{x\to 0} \frac{\sqrt{1+x+x^2}-1}{x}$
[label=(P)]

(a) 1

(b) 0

(c) $\frac{-1}{\sqrt{2}}$

(d) 1/2

**Answer:** (D)
**Strategy (L'Hôpital's Rule):** Plugging in 0 gives 0/0.
Bottom derivative is 1.
Top derivative (chain rule): $\frac{1}{2\sqrt{1+x+x^2}} \cdot (1+2x)$.
Plug in $x = 0$: $\frac{1}{2(1)} \cdot (1+0) = 1/2$.

# PART II: INFINITY LIMITS

28. Evaluate $\lim_{x\to\infty} \left(1 - \frac{2}{x^3}\right)$
[label=()]

(a) -1

(b) $\infty$

(c) 2

(d) 1

**Answer:** (D)
**Strategy (Bottom-Heavy Rule):** Any time you have a constant divided by $x$ going to infinity, that fraction turns into 0. The expression becomes $1 - 0 = 1$.

29. Determine $\lim_{x\to\infty}(2^{-x} + 3)$
[label=()]

(a) 2

(b) 3

(c) -3

(d) 0

**Answer:** (B)
**Strategy (Negative Exponents to Zero):** Rewrite $2^{-x}$ as $1/2^x$. As $x$ grows infinitely large, 1 divided by a massive number becomes 0. The limit is $0 + 3 = 3$.

30. Obtain $\lim_{x\to\infty}\left(\frac{x^2+1}{3x^3-4x+5}\right)$

[label=()]

(a) 3

(b) -4

(c) 5

(d) 0

**Answer:** (D)
**Strategy (Infinity Degree Rule):** Look at the highest powers. The bottom power ($x^3$) is larger than the top power ($x^2$). When the denominator grows faster than the numerator, the entire fraction collapses to 0.

31. Find $\lim_{x\to 2}\left(\frac{x-1}{x^2+x-2}\right)$

[label=()]

(a) 1

(b) 1/2

(c) 1/4

(d) 4

**Answer:** (C)
**Strategy (Direct Substitution Trick):** Don't be fooled into factoring or deriving immediately! Always try plugging the number in first.
Numerator: $2 - 1 = 1$.
Denominator: $4 + 2 - 2 = 4$.
It evaluates cleanly to 1/4.

32. Evaluate $\lim_{x\to -1}\left(\frac{x^2+6x+5}{x^2-2x-3}\right)$

[label=()]

(a) 1

(b) -1

(c) 1/4

(d) -4

**Answer:** (B)
**Strategy (L'Hôpital's Rule):** Plugging in -1 gives 0/0.
Derive top: $2x + 6$
Derive bottom: $2x - 2$
Plug in -1 again: $\frac{-2+6}{-2-2} = \frac{4}{-4} = -1$.

33. Obtain $\lim_{x\to 3}\left(\frac{x^2-9}{x-3}\right)$
[label=()]

(a) 3
(b) -3
(c) 6
(d) 9

**Answer:** (C)
**Strategy (L'Hôpital or Quick Factor):** Plugging in 3 gives 0/0.
*Method 1 (L'Hôpital)*: Derive top ($2x$), derive bottom (1). Plug in 3: $2(3)/1 = 6$.
*Method 2 (Factor)*: The top is a difference of squares $(x-3)(x+3)$. The $(x-3)$ cancels the bottom, leaving just $(x+3)$. Plug in 3 to get $3+3=6$. Both take about 5 seconds!

**Evaluate** $\lim_{x\to 3}\left(\frac{1}{x-3} - \frac{2}{x^2-4x+3}\right)$
(A) 2 (B) -2 (C) 1 (D) $\frac{1}{2}$
**Solution:** Combine the fractions first. $x^2 - 4x + 3$ factors into $(x-3)(x-1)$.

$$\frac{1}{x-3} - \frac{2}{(x-3)(x-1)} = \frac{(x-1)-2}{(x-3)(x-1)} = \frac{x-3}{(x-3)(x-1)} = \frac{1}{x-1}$$

Now substitute $x = 3$: $\frac{1}{3-1} = \mathbf{\frac{1}{2}}$.
**Method Explanation:** We used **algebraic simplification**. When you see two fractions with a shared factor in the denominator resulting in $\infty - \infty$, combining them usually reveals a "hole" that can be canceled out.
**Correct Option: (D)**

**Evaluate** $\lim_{x\to\infty}\left(\frac{2x^3}{x^3+1}\right)$
(A) 0 (B) -1 (C) 2 (D) $\infty$
**Solution:** Since $x$ goes to infinity, compare the highest powers. The numerator has $x^3$ and the denominator has $x^3$. The limit is the ratio of their coefficients: $\frac{2}{1} = \mathbf{2}$.
**Method Explanation: Degree Rule**. For limits at infinity, if the degree of the numerator equals the degree of the denominator, the answer is simply the ratio of the leading coefficients.
**Correct Option: (C)**

**Evaluate** $\lim_{x\to 0} \dfrac{\sqrt{2-x}-\sqrt{2+x}}{x}$
(A) 1 (B) 0 (C) $\frac{-1}{\sqrt{2}}$ (D) $\sqrt{2}$
**Solution:** Use **L'Hôpital's Rule** (differentiate top and bottom).

$$Derivative of top: \frac{-1}{2\sqrt{2-x}} - \frac{1}{2\sqrt{2+x}}$$

Plug in $x = 0$: $\frac{-1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{-2}{2\sqrt{2}} = -\mathbf{\frac{1}{\sqrt{2}}}$.
**Method Explanation:** L'Hôpital's Rule is the fastest way to solve 0/0 forms involving radicals. Differentiating is often faster than multiplying by the conjugate.
**Correct Option: (C)**

**Evaluate** $\lim_{x\to 0} \dfrac{\sqrt{1+x+x^2}-1}{x}$
(A) 1 (B) 0 (C) $\frac{-1}{\sqrt{2}}$ (D) $\frac{1}{2}$
**Solution:** Use **L'Hôpital's Rule**.

$$Derivative of top: \frac{1+2x}{2\sqrt{1+x+x^2}}$$

Plug in $x = 0$: $\frac{1+2(0)}{2\sqrt{1+0+0}} = \mathbf{\frac{1}{2}}$.
**Method Explanation:** Again, for a 0/0 radical limit, the derivative at the point of interest provides the answer instantly.
**Correct Option: (D)**

**Find the exact value of** $\lim_{x\to 0} \dfrac{\sqrt{3+x}-\sqrt{3}}{x}$
(A) $\sqrt{3}$ (B) 0 (C) $\frac{1}{2\sqrt{3}}$ (D) $\sqrt{2}$
**Solution:** Use **L'Hôpital's Rule**.

$$Derivative of top: \frac{1}{2\sqrt{3+x}}$$

Plug in $x = 0$: $\mathbf{\frac{1}{2\sqrt{3}}}$.
**Method Explanation:** This is actually the definition of the derivative of $f(x) = \sqrt{x}$ at $x = 3$. L'Hôpital's gets you there in one step.
**Correct Option: (C)**

**Find the domain of the function** $f(x) = \sqrt{1-x}\ln x$
(A) $(0, +\infty)$ (B) $(0, 1)$ (C) $(-\infty, 1)$ (D) $(0, 1]$
**Solution:** 1. For $\sqrt{1-x}$: $1 - x \geq 0 x \leq 1$.
2. For $\ln x$: $x > 0$.
Combining these gives $0 < x \leq 1$, which is $(0, 1]$.

**Method Explanation:** To find the domain, you must satisfy all restrictions simultaneously (radicals must be $\geq 0$ and logs must be $> 0$).
**Correct Option: (D)**

**Evaluate** $\lim_{x \to -1} \left\{ \dfrac{x^2 - x + 4}{3x} \right\}$
(A) 2 (B) -2 (C) 5 (D) 0
**Solution:** Use **Direct Substitution**.

$$\frac{(-1)^2 - (-1) + 4}{3(-1)} = \frac{1 + 1 + 4}{-3} = \frac{6}{-3} = \mathbf{-2}$$

**Method Explanation:** Always try plugging in the value first. If the denominator isn't zero, the result of the substitution is your limit.
**Correct Option: (B)**

**Evaluate** $\lim_{x \to 0} \left\{ \dfrac{\sin x}{\tan x} \right\}$
(A) 1 (B) 2 (C) 0 (D) $\infty$
**Solution:** Simplify the trig expression.

$$\frac{\sin x}{\sin x / \cos x} = \cos x$$

Substitute $x = 0$: $\cos(0) = \mathbf{1}$.
**Method Explanation: Trig simplification**. Before using L'Hôpital's, check if the trig functions cancel out. It is often faster.
**Correct Option: (A)**

**Find** $\lim_{x \to 3} \dfrac{x^2 - x - 6}{x - 3}$
A. 15 B. 3 C. 5 D. -5 E. 0
**Solution:** Use **L'Hôpital's Rule**.

$$Derivative: \frac{2x - 1}{1}$$

Plug in $x = 3$: $2(3) - 1 = \mathbf{5}$.
**Method Explanation:** For 0/0 rational functions, differentiating is usually faster than factoring.
**Correct Option: (C)**

**Determine** $\lim_{x \to 3} (x^2 - 2)^2$
A. 45 B. 33 C. 2 D. 7 E. 49
**Solution: Direct Substitution**.

$$(3^2 - 2)^2 = (9 - 2)^2 = 7^2 = \mathbf{49}$$

**Correct Option: (E)**

**Evaluate** $\lim_{x\to 2} \dfrac{x^2+6x+5}{x^2-2x-3}$
A. -15 B. 30 C. 67 D. -7 E. 18
**Solution: Direct Substitution**.

$$\frac{2^2+6(2)+5}{2^2-2(2)-3} = \frac{4+12+5}{4-4-3} = \frac{21}{-3} = -\mathbf{7}$$

**Method Explanation:** Since the denominator is not zero at $x = 2$, substitution works perfectly.
**Correct Option: (D)**

**Obtain** $\lim_{x\to -1} \dfrac{x^2+6x+5}{x^2-2x-3}$
A. 15 B. -3 C. 5 D. -1 E. 9
**Solution:** This yields 0/0. Use **L'Hôpital's Rule**.

$$Derivative: \frac{2x+6}{2x-2}$$

Plug in $x = -1$: $\frac{2(-1)+6}{2(-1)-2} = \frac{4}{-4} = -\mathbf{1}$.
**Correct Option: (D)**

**Evaluate** $\lim_{x\to\infty} \left\{\dfrac{1}{x^3}\right\}$
(A) 0 (B) $\infty$ (C) 2 (D) 1
**Solution: Degree Rule**. Since the degree of the denominator (3) is greater than the degree of the numerator (0), the limit is **0**.
**Correct Option: (A)**

**Evaluate** $\lim_{x\to 0} \left\{\dfrac{1}{x^2}\right\}$
(A) 0 (B) $\infty$ (C) 2 (D) 1
**Solution:** As $x$ approaches 0, $x^2$ becomes a very small positive number. $1/(verysmallnumber) = \infty$.
**Correct Option: (B)**

**Evaluate** $\lim_{x\to\infty} \left\{\dfrac{2x^3}{x^3+1}\right\}$
(A) -1 (B) $\infty$ (C) 2 (D) 0
**Solution:** Identical to Q13. Same degree ratio of coefficients = **2**.
**Correct Option: (C)**

**Evaluate** $\lim_{x\to\infty}\left(1-\frac{2}{x^3}\right)$
(A) -1 (B) $\infty$ (C) 2 (D) 1
**Solution:** As $x\to\infty$, $\frac{2}{x^3}\to 0$. So, $1-0=\mathbf{1}$.
**Correct Option: (D)**

**Determine** $\lim_{x\to\infty}(2^{-x}+3)$
(A) 2 (B) 3 (C) -3 (D) 0
**Solution:** $2^{-x}$ is the same as $\frac{1}{2^x}$. As $x\to\infty$, $\frac{1}{2^x}\to 0$. So, $0+3=\mathbf{3}$.
**Correct Option: (B)**

**Obtain** $\lim_{x\to\infty}\left(\frac{x^2+1}{3x^3-4x+5}\right)$
(A) 3 (B) -4 (C) 5 (D) 0
**Solution: Degree Rule**. The degree of the denominator (3) is higher than the numerator (2). Thus, the limit is **0**.
**Correct Option: (D)**

**Find** $\lim_{x\to 2}\left(\frac{x-1}{x^2+x-2}\right)$
(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4
**Solution: Direct Substitution**.

$$\frac{2-1}{2^2+2-2}=\frac{1}{4}$$

**Correct Option: (C)**

**Evaluate** $\lim_{x\to -1}\left(\frac{x^2+6x+5}{x^2-2x-3}\right)$
(A) 1 (B) -1 (C) $\frac{1}{4}$ (D) -4
**Solution:** (Duplicate of Q24). Result is **-1**.
**Correct Option: (B)**

**Obtain** $\lim_{x\to 3}\left(\frac{x^2-9}{x-3}\right)$
(A) 3 (B) -3 (C) 6 (D) 9
**Solution:** Use **L'Hôpital's Rule**.

$$Derivative: \frac{2x}{1}$$

Plug in $x=3$: $2(3)=\mathbf{6}$.
**Correct Option: (C)**

## 34. Find $\lim_{h\to 0} \frac{|h|}{h}$

**Options:** (A) 1 (B) -1 (C) $\infty$ **(D) The limit does not exist**
**Solution (Short Trick - Numerical Check):** Check values very close to 0 from both sides:

- From the right ($h = 0.01$): $\frac{0.01}{0.01} = 1$
- From the left ($h = -0.01$): $\frac{|-0.01|}{-0.01} = \frac{0.01}{-0.01} = -1$

Since $1 \neq -1$, the limit **DNE**.

## 35. Find the value of $\lim_{x\to\infty} \frac{x+2}{9x^2+1}$

**Options: (A) 0** (B) 1/9 (C) 2/9 (D) $\infty$
**Solution (Infinity Rule):** When $x \to \infty$, if the **degree of the denominator (2)** is greater than the **degree of the numerator (1)**, the limit is always **0**.

## 36. Find $\lim_{x\to 2} \frac{3x^2-12}{x-2}$

**Options:** (A) 0 **(B) 12** (C) -12 (D) 1
**Solution (L'Hôpital's Rule):** Substitution gives $\frac{0}{0}$. Differentiate top and bottom:

$$\lim_{x\to 2} \frac{\frac{d}{dx}(3x^2 - 12)}{\frac{d}{dx}(x - 2)} = \lim_{x\to 2} \frac{6x}{1} = 6(2) = 12$$

## 37. Find $\lim_{n\to\infty} \left[\frac{3^{n+1}-5^{n+1}}{3^n-5^n}\right]$

**Options:** (A) -5 **(B) 5** (C) -12 (D) 1
**Solution (Dominant Term Trick):** At infinity, the largest base ($5^n$) dominates. Ignore the $3^n$ terms:

$$\lim_{n\to\infty} \frac{-5^{n+1}}{-5^n} = \frac{-5 \cdot 5^n}{-5^n} = 5$$

## 38. Find $\lim_{x\to 4} \left[\frac{x-4}{\sqrt{x}-2}\right]$

**Options:** (A) -5 (B) 5 **(C) 4** (D) 1
**Solution (L'Hôpital's Rule):**

$$\lim_{x\to 4} \frac{1}{\frac{1}{2\sqrt{x}}} = \lim_{x\to 4} 2\sqrt{x} = 2\sqrt{4} = 4$$

**39. Find** $\lim_{x\to 0} \frac{2\tan x - x\cos x}{x+\sin x}$

**Options:** (A) -5 **(B)** 1/2 (C) $-1/2$ (D) 1
**Solution (L'Hôpital's Rule):** Differentiate numerator and denominator:

$$\lim_{x\to 0} \frac{2\sec^2 x - (\cos x - x\sin x)}{1+\cos x} = \frac{2(1)^2 - (1-0)}{1+1} = \frac{1}{2}$$

**42. Solve using L'Hôpital's Rule:** $\lim_{x\to 1} \frac{e^{x-1}-x}{(x-1)^2}$

**Solution:** First derivative $(\frac{0}{0})$: $\frac{e^{x-1}-1}{2(x-1)}$. Second derivative: $\frac{e^{x-1}}{2}$.

$$At x = 1 : \frac{e^0}{2} = \frac{1}{2}$$

# Section B: Theory (Short Solutions)

**i.** $\lim_{x\to\infty} \frac{6x^2+7x-3}{20x^2+13x-15}$

**Trick:** Degrees are equal ($x^2$). Take the ratio of leading coefficients: $\frac{6}{20} = \mathbf{\frac{3}{10}}$.

**ii.** $\lim_{x\to 0} \frac{x}{\sqrt{x+9}-3}$

**Trick (L'Hôpital):** $\frac{1}{\frac{1}{2\sqrt{x+9}}} = 2\sqrt{x+9}$. At $x=0$, $2\sqrt{9} = \mathbf{6}$.

**iv.** $\lim_{x\to\infty} \frac{-x^3+2x^2-x+1}{3+3x-4x^2-8x^3}$

**Trick:** Highest power is $x^3$ on top and bottom. Limit is $\frac{-1}{-8} = \mathbf{\frac{1}{8}}$.

# Part II: Differentiation and Applications

## Question 46

**Question:** Let $f(x) = \frac{5}{2}x^2 - e^x$, find the value of $x$ for which the second derivative $f''(x)$ equals zero.
**Options:**
(A) $5e$
(B) 0
(C) $\ln 5$
(D) $e^5$
**Solution:**

- **Step 1: Find the first derivative, $f'(x)$.**
  Using the Power Rule and Exponential Rule:
$$f'(x) = \frac{d}{dx}\left(\frac{5}{2}x^2\right) - \frac{d}{dx}(e^x) = 5x - e^x$$
- **Step 2: Find the second derivative, $f''(x)$.**
  Differentiate $f'(x)$:
$$f''(x) = \frac{d}{dx}(5x) - \frac{d}{dx}(e^x) = 5 - e^x$$
- **Step 3: Solve $f''(x) = 0$.**
$$5 - e^x = 0 e^x = 5$$
  Taking the natural log of both sides:
$$x = \ln 5$$

**Answer: (C)**

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## Question 47

**Question:** Find the derivative of $f(x) = \left(1 + x^4 - \frac{1}{x}\right)^{\frac{5}{3}}$.
**Options:**
(A) $\frac{20}{3}x^{\frac{17}{3}} + \frac{5}{3x^{\frac{8}{3}}}$
(B) $\frac{5}{3}\left(1 + x^4 + \frac{1}{x}\right)^{\frac{2}{3}}$
(C) $\frac{5}{3}\left(1 + x^4 - \frac{1}{x}\right)^{\frac{2}{3}}\left(4x^3 + \frac{1}{x^2}\right)$
(D) $\frac{5}{3}x^{\frac{2}{3}}\left(4x^3 + \frac{1}{x^2}\right)$
**Solution:**

- **Step 1: Rewrite the function.**
$$f(x) = (1 + x^4 - x^{-1})^{\frac{5}{3}}$$
- **Step 2: Apply the Chain Rule.**
  Let $u = 1 + x^4 - x^{-1}$. Then $f(x) = u^{5/3}$.
$$f'(x) = \frac{5}{3}u^{\frac{5}{3}-1} \cdot \frac{du}{dx}$$
$$f'(x) = \frac{5}{3}(1 + x^4 - x^{-1})^{\frac{2}{3}} \cdot (4x^3 - (-x^{-2}))$$
- **Step 3: Simplify.**
$$f'(x) = \frac{5}{3}\left(1 + x^4 - \frac{1}{x}\right)^{\frac{2}{3}}\left(4x^3 + \frac{1}{x^2}\right)$$

**Answer: (C)**

---

## Question 48

**Question:** Differentiate $y = \frac{x^2+1}{3x^2+2}$ using quotient rule.
**Options:**
(A) $\frac{2x}{(3x^2+2)^2}$
(B) $\frac{-2x}{(3x^2+2)^2}$
(C) $\frac{2x}{(4x^2+2)^3}$
(D) $\frac{3x}{(3x^2+2)^2}$
**Solution:**

- **Step 1: Identify $u$ and $v$.**
  $u = x^2 + 1 u' = 2x$
  $v = 3x^2 + 2 v' = 6x$
- **Step 2: Apply the Quotient Rule $\frac{vu'-uv'}{v^2}$.**

$$y' = \frac{(3x^2+2)(2x) - (x^2+1)(6x)}{(3x^2+2)^2}$$

- **Step 3: Simplify the numerator.**

$$Numerator = 6x^3 + 4x - (6x^3 + 6x) = -2x$$

$$y' = \frac{-2x}{(3x^2+2)^2}$$

**Answer: (B)**

## Question 49

**Question:** Differentiate $y = \sin(2x+1)$.
**Solution:**

- **Step 1: Apply the Chain Rule.**
  The derivative of $\sin(u)$ is $\cos(u) \cdot u'$.

$$y' = \cos(2x+1) \cdot \frac{d}{dx}(2x+1)$$

$$y' = \cos(2x+1) \cdot 2 = 2\cos(2x+1)$$

**Answer: Option (B) from the following list.**

## Question 51

**Question:** Differentiate $x^3 + 3xy^2 - 2y^3 = 3$ with respect to $x$.
**Options:**
(B) $\frac{-3(x^2+y^2)}{6y(x-y)}$
**Solution:**

- **Step 1: Implicit Differentiation.**

$$\frac{d}{dx}(x^3) + \frac{d}{dx}(3xy^2) - \frac{d}{dx}(2y^3) = 0$$

- **Step 2: Apply Product Rule for $3xy^2$.**

$$3x^2 + [3y^2 + 3x(2yy')] - 6y^2y' = 0$$
$$3x^2 + 3y^2 + 6xyy' - 6y^2y' = 0$$

- **Step 3: Solve for $y'$.**

$$y'(6xy - 6y^2) = -3x^2 - 3y^2$$
$$y' = \frac{-3(x^2 + y^2)}{6y(x - y)}$$

**Answer: (B)**

---

## Question 53

**Question:** Differentiate $4\cos x \sin y + 2x \cos y = 3y + 2x - 5$.
**Solution:**

- **Step 1: Differentiate using Product Rule.**
  $\frac{d}{dx}(4\cos x \sin y) = -4\sin x \sin y + 4\cos x \cos y \cdot y'$
  $\frac{d}{dx}(2x\cos y) = 2\cos y - 2x \sin y \cdot y'$
- **Step 2: Group $y'$ terms.**

$$y'(4\cos x \cos y - 2x \sin y - 3) = 2 + 4\sin x \sin y - 2\cos y$$
$$y' = \frac{2 + 4\sin y \sin x - 2\cos y}{4\cos x \cos y - 2x \sin y - 3}$$

**Answer: (C)**

---

## Question 54

**Question:** Differentiate $y = \sin^{-1} x$.
**Answer:**
Using the standard derivative formula:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

**Answer: (B)**

## 55. Differentiate $y = \cos^{-1} x$

(A) $\frac{1}{1-e^2}$ (B) $\frac{1}{\sqrt{1-x^2}}$ (C) $\frac{1}{1+x^2}$ (D) $\frac{1}{\sqrt{1+x^2}}$

**Solution:** The standard derivative for the inverse cosine function is:

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

*Note: None of the provided options include the negative sign. Option (B) is the derivative of* $\sin^{-1} x$*, but it is the closest match provided in the list.*

## 56. Which of the following statements are true?

(A) If $f$ is continuous at $a$, then $f$ is differentiable at $a$.
(B) If $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ exists, then $f$ is differentiable at $a$.
(C) $\lim_{x\to a} f(x)$ exists, then $f$ is differentiable at $a$.
(D) All of the above.

**Solution:**

1. Statement (A) is false; continuity does not guarantee differentiability (e.g., $f(x) = |x|$ at $x = 0$).
2. Statement (B) is the formal definition of the derivative. If this limit exists, the function is differentiable by definition.
3. Statement (C) is false; a limit existing at a point doesn't even guarantee continuity, let alone differentiability.

**Correct Option: (B)**

## 57. If $F(x) = f(g(x))$, where $f(-2) = 8, f'(-2) = 4, f'(5) = 3, g(5) = -2, g'(5) = 6$, find $F'(5)$

(A) 24 (B) 8 (C) 12 (D) 20

**Solution:** Using the **Chain Rule**: $F'(x) = f'(g(x)) \cdot g'(x)$.

1. $F'(5) = f'(g(5)) \cdot g'(5)$
2. Substitute $g(5) = -2$ and $g'(5) = 6$: $F'(5) = f'(-2) \cdot 6$
3. Substitute $f'(-2) = 4$: $F'(5) = 4 \cdot 6 = 24$

**Correct Option: (A)**

## 58. Find the Gradient of $(x^2+1)^8$ at the point $x=1$

(A) $2\times10^2$ (B) $2.033\times10^4$ (C) $2.048\times10^3$ (D) 2015

**Solution:** Apply the **Power Chain Rule**: $\frac{d}{dx}[u^n] = n\cdot u^{n-1}\cdot u'$.

1. Let $y=(x^2+1)^8$.
2. $y' = 8(x^2+1)^7\cdot(2x) = 16x(x^2+1)^7$.
3. At $x=1$: $y' = 16(1)(1^2+1)^7 = 16(2)^7$.
4. $16\cdot128 = 2048 = 2.048\times10^3$.

**Correct Option: (C)**

## 59. Find $\frac{dy}{dx}$ in terms of $x$ and $y$ if $x^2+y^3=xy$

(A) $\frac{y-2x}{3y^2-x}$ (B) $\frac{3y-2x}{3y^3-2x}$ (C) $\frac{2y-3x}{3y^2-x}$ (D) $\frac{y-2x}{3y^3-3x}$

**Solution:** Use **Implicit Differentiation**:

1. Differentiate both sides with respect to $x$: $2x+3y^2\frac{dy}{dx} = y+x\frac{dy}{dx}$
2. Group $\frac{dy}{dx}$ terms: $3y^2\frac{dy}{dx} - x\frac{dy}{dx} = y-2x$
3. Factor out $\frac{dy}{dx}$: $\frac{dy}{dx}(3y^2-x) = y-2x$
4. Solve: $\frac{dy}{dx} = \frac{y-2x}{3y^2-x}$

**Correct Option: (A)**

# Page 18: Further Differentiation

## 60. Differentiate $y = x^2 \sin x$

(A) $x^2 + \sin x$ (B) $x^2 \cos x + \sin x$ (C) $2x + \cos x$ (D) $x^2 \cos x + 2x \sin x$

**Solution:** Use the **Product Rule**: $(uv)' = u'v + uv'$.

1. Let $u = x^2$ and $v = \sin x$.
2. $u' = 2x$, $v' = \cos x$.
3. $y' = (2x)(\sin x) + (x^2)(\cos x)$.

**Correct Option: (D)**

## 61. Find $\frac{d^2y}{dx^2}$, if $y = x^3 + \cos x$

(A) $3x^2 - \sin x$ (B) $6x - \cos x$ (C) $3x^3 + \cos x$ (D) $3x \sin x$

**Solution:**

1. First derivative: $\frac{dy}{dx} = 3x^2 - \sin x$.
2. Second derivative: $\frac{d^2y}{dx^2} = \frac{d}{dx}(3x^2) - \frac{d}{dx}(\sin x) = 6x - \cos x$.

**Correct Option: (B)**

## 62. The gradient of the curve $y = -2x^3 + 3x + 5$ at $x = 2$ is

(A) -21 (B) 27 (C) -16 (D) -5

**Solution:** The gradient is the first derivative $y'$.

1. $y' = -6x^2 + 3$.
2. At $x = 2$: $y' = -6(2^2) + 3 = -6(4) + 3 = -24 + 3 = -21$.

**Correct Option: (A)**

## 64. The first derivative of $f(x) = 2x(x^2 - 5x + 7)$ at $x = 2$ is

(A) 1 (B) 3 (C) -2 (D) -1

**Solution:**

1. Simplify $f(x) = 2x^3 - 10x^2 + 14x$.
2. $f'(x) = 6x^2 - 20x + 14$.
3. At $x = 2$: $f'(2) = 6(4) - 20(2) + 14 = 24 - 40 + 14 = -2$.

**Correct Option: (C)**

**65. If $f(x) = 5x^3 - 2x^2 + 6x + 1$, then $f'''(x)$ is**

(A) 30 (B) 10 (C) 40 (D) 50

**Solution:**

1. $f'(x) = 15x^2 - 4x + 6$
2. $f''(x) = 30x - 4$
3. $f'''(x) = 30$

**Correct Option: (A)**

**66. If $y(x) = \tan x - x$, then $y'(x)$ is**

(A) $\sec^2 x$ (B) $\sec^2 x + 1$ (C) $\csc^2 x - 1$ (D) $\tan^2 x$

**Solution:**

1. $y' = \sec^2 x - 1$.
2. Using the identity $1 + \tan^2 x = \sec^2 x$, we know $\sec^2 x - 1 = \tan^2 x$.

**Correct Option: (D)**

**67. If $y(x) = \frac{1}{\sqrt{x}}$, then $y'(x)$ is**

(A) $\frac{1}{2x\sqrt{x}}$ (B) $\frac{-1}{2\sqrt{x}}$ (C) $\frac{1}{\sqrt{x}}$ (D) $\frac{-1}{2x\sqrt{x}}$

**Solution:**

1. Rewrite $y = x^{-1/2}$.
2. $y' = -\frac{1}{2}x^{-3/2} = -\frac{1}{2x^{3/2}} = -\frac{1}{2x\sqrt{x}}$.

**Correct Option: (D)**

## Page 19: Applications and Basics

**68. For what value(s) of $x$ does the graph of $f(x) = \frac{1}{3}x^3 - x^2 + 3$ have a horizontal tangent?**

(A) 0 (B) 0 and 3 (C) 2 (D) 0 and 2

**Solution:** Horizontal tangents occur where $f'(x) = 0$.

1. $f'(x) = x^2 - 2x$.
2. Set to zero: $x(x-2) = 0$.
3. $x = 0, x = 2$.

**Correct Option: (D)**

**69. Given the function $f(x) = x^3 - 4x^2 + 5x$, find the open interval(s) where $f$ is concave down, i.e. where $f''(x) < 0$**

(A) $(\frac{4}{3}, +\infty)$ (B) $(-\infty, \frac{4}{3})$ (C) $(\frac{3}{4}, +\infty)$ (D) $(-\infty, \frac{3}{4})$

**Solution:**

1. $f'(x) = 3x^2 - 8x + 5$
2. $f''(x) = 6x - 8$
3. Set $f''(x) < 0 \Rightarrow 6x - 8 < 0 \Rightarrow 6x < 8 \Rightarrow x < \frac{4}{3}$.

**Correct Option: (B)**

**70. Differentiate $y = \frac{3}{x} + 2$**

(A) $-\frac{3}{x^2} + 2x$ (B) $\frac{3}{x^2}$ (C) $-\frac{3}{x^2}$ (D) $-\frac{3}{x^3}$

**Solution:**

1. Rewrite $y = 3x^{-1} + 2$.
2. $y' = -3x^{-2} = -\frac{3}{x^2}$.

**Correct Option: (C)**

**71. Differentiate $y = 4x^3 + 4x + 10$**

(A) $2x$ (B) $12x^2 + 4x$ (C) $12x^2 + 4$ (D) $8x + 2$

**Solution:** Using the power rule: $y' = 3(4)x^2 + 4 = 12x^2 + 4$. **Correct Option: (C)**

**72. Differentiate $y = \tan x$**

(A) $\cos^2 x$ (B) $\sec^2 x$ (C) $-\sin x$ (D) $\frac{1}{\sec^2 x}$

**Solution:** This is a fundamental derivative: $\frac{d}{dx}(\tan x) = \sec^2 x$. **Correct Option: (B)**

**73. Differentiate** $y = 5e^{3x}$

(A) $5e^{3x}$ (B) $8x^{3x} + 4x$ (C) $15e^{3x}$ (D) $5e^{2x}$

**Solution:** Using the chain rule for exponentials: $\frac{d}{dx}[e^{kx}] = ke^{kx}$.

1. $y' = 5 \cdot (3e^{3x}) = 15e^{3x}$.

**Correct Option: (C)**

**74. Differentiate** $y = 5\sin x + 3\cos x + 5x^2$

(A) $3\sin x - 5\cos x - 10x$

(B) $5\cos x + 4x$

(C) $5\cos x + 3\sin x + 10x$

(D) $5\cos x - 3\sin x + 10x$

**Solution:**

- **Rules Used:** Sum rule (differentiate each term separately), Power Rule ($\frac{d}{dx}x^n = nx^{n-1}$), and standard trigonometric derivatives ($\frac{d}{dx}\sin x = \cos x$, $\frac{d}{dx}\cos x = -\sin x$).
- **Step 1:** Differentiate the first term: $\frac{d}{dx}(5\sin x) = 5\cos x$
- **Step 2:** Differentiate the second term: $\frac{d}{dx}(3\cos x) = 3(-\sin x) = -3\sin x$
- **Step 3:** Differentiate the third term: $\frac{d}{dx}(5x^2) = 5(2)x^{2-1} = 10x$
- **Step 4:** Combine the results: $\frac{dy}{dx} = 5\cos x - 3\sin x + 10x$

**Correct Option: (D)**

---

**75. Differentiate** $x^6 - 7x^3 - 6x + 4$

(A) $6x^5 - 21x^2 - 7$

(B) $6x^5 - 21x^2 - 6$

(C) $6x^6 - 21x^3 - 6x$

(D) $6x^5 - 21x^2 - 6$ *(Note: Options B and D appear identical in the original image)*

**Solution:**

- **Rule Used:** Power Rule ($\frac{d}{dx}x^n = nx^{n-1}$) and the derivative of a constant is 0.
- **Step 1:** $\frac{d}{dx}(x^6) = 6x^5$

- **Step 2:** $\frac{d}{dx}(-7x^3) = -7(3)x^2 = -21x^2$
- **Step 3:** $\frac{d}{dx}(-6x) = -6$
- **Step 4:** $\frac{d}{dx}(4) = 0$
- **Step 5:** Combine the results: $6x^5 - 21x^2 - 6$

**Correct Option: (B) or (D)**

---

**76. Differentiate $y = x^2 \sin x$ using product rule**

(A) $2x \cos x$

(B) $2x^2 \cos x$

(C) $x^2 \cos x + 2x \sin x$

(D) $x^2 \sin x + 2x \cos x$

**Solution:**

- **Rule Used:** Product Rule: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
- **Step 1:** Identify $u$ and $v$. Let $u = x^2$ and $v = \sin x$.
- **Step 2:** Find the derivatives: $u' = 2x$ and $v' = \cos x$.
- **Step 3:** Apply the formula:

$$y' = (x^2)(\cos x) + (\sin x)(2x)$$

- **Step 4:** Rearrange: $y' = x^2 \cos x + 2x \sin x$

**Correct Option: (C)**

---

**77. Differentiate $y = 6x^2 \sin x \cos x$**

(A) $6x^2 \cos 2x + 6x(2 \cos x \sin x)$

(B) $12x^2 - \cos x + \cos x$

(C) $6x^2 \sin x + 2(3 \cos x \sin x)$

(D) 1

**Solution:**

- **Rules Used:** Product Rule and Double Angle Identity ($\sin 2x = 2 \sin x \cos x$).

- **Step 1:** Simplify the function first using the double angle identity.

$$y = 3x^2(2\sin x \cos x) = 3x^2 \sin(2x)$$

- **Step 2:** Apply the product rule where $u = 3x^2$ and $v = \sin(2x)$.

$$u' = 6x \quad and \quad v' = 2\cos(2x) \quad (usingchainrule)$$

- **Step 3:** Apply the formula $\frac{dy}{dx} = uv' + vu'$:

$$y' = 3x^2(2\cos 2x) + \sin(2x)(6x)$$

$$y' = 6x^2 \cos 2x + 6x \sin 2x$$

- **Step 4:** Expand $\sin 2x$ back to $2\sin x \cos x$ to match the format of Option A:

$$y' = 6x^2 \cos 2x + 6x(2\sin x \cos x)$$

**Correct Option: (A)** *(Note: The transcription of Option A includes an 'x' before the parenthesis, which is slightly blurred in the image but is mathematically the only way the answer works).*

---

**78. Differentiate $y = (x^2 + 2x + 1)(x^3 + 3x)$ using product rule**

(A) $5x^4 + 8x^3 + 12x^2 + 12x + 3$

(B) $6x^5 + 8x^3 + 12x^2 + 12x + 3$

(C) $5x^5 + 8x^4 + 12x^3 + 12x + 3$

(D) $5x^4 + 8x^3 + 12x^2 + 12x$

**Solution:**

- **Rule Used:** Product Rule: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
- **Step 1:** Let $u = x^2 + 2x + 1$ and $v = x^3 + 3x$.
- **Step 2:** Find the derivatives: $u' = 2x + 2$ and $v' = 3x^2 + 3$.
- **Step 3:** Apply the formula:

$$y' = (x^2 + 2x + 1)(3x^2 + 3) + (x^3 + 3x)(2x + 2)$$

- **Step 4:** Expand both sides:
  Left part: $(3x^4 + 3x^2 + 6x^3 + 6x + 3x^2 + 3) = 3x^4 + 6x^3 + 6x^2 + 6x + 3$
  Right part: $(2x^4 + 2x^3 + 6x^2 + 6x)$

- **Step 5:** Add them together:

$$(3x^4 + 2x^4) + (6x^3 + 2x^3) + (6x^2 + 6x^2) + (6x + 6x) + 3$$

$$= 5x^4 + 8x^3 + 12x^2 + 12x + 3$$

**Correct Option: (A)**

---

**79. Differentiate** $y = \frac{1}{\sqrt{x}}$

(A) $-\frac{1}{2}x^{-\frac{3}{2}}$

(B) $\frac{1}{\sqrt{x}}$

(C) $e^x$

(D) $\frac{1}{x^2}$

**Solution:**

- **Rule Used:** Power Rule. First, convert the radical into an exponent.
- **Step 1:** Rewrite the function: $y = x^{-\frac{1}{2}}$
- **Step 2:** Apply the Power Rule ($\frac{d}{dx}x^n = nx^{n-1}$):

$$y' = -\frac{1}{2}x^{-\frac{1}{2}-1}$$

$$y' = -\frac{1}{2}x^{-\frac{3}{2}}$$

**Correct Option: (A)**

---

**80. Determine the co-ordinates of the point on the graph** $y = 3x^2 - 7x + 2$ **where the gradient is** 1**.**
*(Note: Based on the options, the gradient in the original question was likely intended to be* $-1$*. I will solve for* $-1$ *to match the provided choices).*

(A) $(4, -2)$

(B) $(1, -2)$

(C) $(-1, -9)$

(D) $(1, 23)$

(E) $(1, 2)$

**Solution:**

- **Rule Used:** The gradient of a curve is its first derivative ($\frac{dy}{dx}$).
- **Step 1:** Find the derivative of the curve:

$$\frac{dy}{dx} = 6x - 7$$

- **Step 2:** Set the gradient to $-1$ (assuming the typo) and solve for $x$:

$$6x - 7 = -1$$

$$6x = 6x = 1$$

- **Step 3:** Substitute $x = 1$ back into the original equation to find the $y$-coordinate:

$$y = 3(1)^2 - 7(1) + 2$$

$$y = 3 - 7 + 2 = -2$$

- **Step 4:** The coordinates are $(1, -2)$.
  *(If we strictly used a gradient of* $1$*,* $x = \frac{4}{3}$*, which yields* $y = -2$*, resulting in* $\left(\frac{4}{3}, -2\right)$ *which is not an option).*

**Correct Option: (B)**

---

**81. Find the gradient of the curve $y = 2t^4 + 3t^3 - t + 4$ at the point $(1, 8)$.**

(A) 107

(B) 267

(C) 12

(D) 16

(E) 8

**Solution:**

- **Rule Used:** Power Rule and evaluating the derivative at a specific point.
- **Step 1:** Find the derivative $\frac{dy}{dt}$:

$$\frac{dy}{dt} = 8t^3 + 9t^2 - 1$$

- **Step 2:** Substitute the $t$-coordinate from the given point ($t = 1$):

$$\left.\frac{dy}{dt}\right|_{t=1} = 8(1)^3 + 9(1)^2 - 1$$

$$= 8 + 9 - 1 = 16$$

**Correct Option: (D)**

---

**82. Differentiate with respect to $t$ the function $y = 5t^4 + 4t^2 - \frac{1}{2t^2} + \frac{1}{\sqrt{t}} - 8$ given that $t = 1$.**

(A) 15

(B) $8\frac{1}{2}$

(C) $18\frac{1}{3}$

(D) $28\frac{1}{2}$

(E) 25

**Solution:**

- **Rule Used:** Power Rule. Convert all fractions and roots to negative/fractional exponents first.
- **Step 1:** Rewrite the function:

$$y = 5t^4 + 4t^2 - \frac{1}{2}t^{-2} + t^{-\frac{1}{2}} - 8$$

- **Step 2:** Differentiate term by term:

$$\frac{dy}{dt} = 20t^3 + 8t - \frac{1}{2}(-2)t^{-3} + \left(-\frac{1}{2}\right)t^{-\frac{3}{2}}$$

$$\frac{dy}{dt} = 20t^3 + 8t + t^{-3} - \frac{1}{2}t^{-\frac{3}{2}}$$

- **Step 3:** Substitute $t = 1$:

$$20(1)^3 + 8(1) + (1)^{-3} - \frac{1}{2}(1)^{-\frac{3}{2}}$$

$$= 20 + 8 + 1 - 0.5$$

$$= 28.5 = 28\frac{1}{2}$$

**Correct Option: (D)**

---

**83. If** $y = \frac{x^3-1}{x}$, $y''(4)$ **is**

(A) $\frac{65}{32}$

(B) $\frac{5}{2}$

(C) $\frac{5}{3}$

(D) $\frac{6}{32}$

**Solution:**

- **Rule Used:** Power Rule. First, split the fraction to simplify. *(Note: To get the exact answer in option A, the original question likely intended a + sign. I will note the calculation for both).*
- **Step 1:** Simplify the equation:

$$y = \frac{x^3}{x} - \frac{1}{x} = x^2 - x^{-1}$$

- **Step 2:** Find the first derivative $y'$:

$$y' = 2x - (-1)x^{-2} = 2x + x^{-2}$$

- **Step 3:** Find the second derivative $y''$:

$$y'' = 2 + (-2)x^{-3} = 2 - \frac{2}{x^3}$$

- **Step 4:** Substitute $x = 4$:

$$y''(4) = 2 - \frac{2}{4^3} = 2 - \frac{2}{64} = 2 - \frac{1}{32}$$

$$y''(4) = \frac{64}{32} - \frac{1}{32} = \frac{63}{32}$$

- *(Note: If the original equation was strictly* $y = \frac{x^3+1}{x}$*, the second derivative becomes* $y'' = 2 + \frac{2}{x^3}$*, which equals* $2 + \frac{1}{32} = \frac{65}{32}$*. Assuming this common typo, the answer maps to Option A).*

**Likely Intended Option: (A)**

# Question 1: Differentiation from First Principles

**Problem:** Obtain the first derivative of $y$ using the first principle given that $y = \frac{1}{\sqrt{x}}$.

**Solution:** The First Principle of Differentiation is defined as:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Given $f(x) = \frac{1}{\sqrt{x}}$: $\mathrm{d}y}{dx} = \lim_{h \to 0} \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h} = \lim_{h \to 0} \frac{\sqrt{x} - \sqrt{x+h}}{h(\sqrt{x}\sqrt{x+h})}$

To solve the limit, we rationalize the numerator by multiplying with the conjugate: $\mathrm{d}y}{dx} = \lim_{h \to 0} \frac{\sqrt{x} - \sqrt{x+h}}{h(\sqrt{x}\sqrt{x+h})} \cdot \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} = \lim_{h \to 0} \frac{x - (x+h)}{h(\sqrt{x}\sqrt{x+h})(\sqrt{x} + \sqrt{x+h})} = \lim_{h \to 0} \frac{-h}{h(\sqrt{x}\sqrt{x+h})(\sqrt{x} + \sqrt{x+h})}$

Canceling $h$ and applying the limit $h \to 0$: $\mathrm{d}y}{dx} = \frac{-1}{(\sqrt{x}\sqrt{x})(\sqrt{x} + \sqrt{x})} = \frac{-1}{x(2\sqrt{x})} = -\frac{1}{2x^{3/2}}$

—

# Question 2: Implicit Differentiation

**Problem:** Given $4x^2 + 2xy^3 - 5y^2 = 0$, obtain the first and second derivatives and evaluate when $x = 1$ and $y = 2$.

**Solution:** Differentiating implicitly with respect to $x$:

$$8x + (2y^3 + 2x \cdot 3y^2 \frac{dy}{dx}) - 10y\frac{dy}{dx} = 0$$

$$8x + 2y^3 + \frac{dy}{dx}(6xy^2 - 10y) = 0$$

$$\frac{dy}{dx} = \frac{-(8x + 2y^3)}{6xy^2 - 10y} = \frac{-(4x + y^3)}{3xy^2 - 5y}$$

Evaluating $\frac{dy}{dx}$ at $(1, 2)$:

$$y' = \frac{-(4(1) + 2^3)}{3(1)(2^2) - 5(2)} = \frac{-(4+8)}{12 - 10} = \frac{-12}{2} = -6$$

For the second derivative, we differentiate $y'(3xy^2 - 5y) = -4x - y^3$:

$$y''(3xy^2 - 5y) + y'(3y^2 + 6xyy' - 5y') = -4 - 3y^2y'$$

Evaluating at $(1, 2)$ with $y' = -6$:

$$y''(2) + (-6)[12 + 6(1)(2)(-6) - 5(-6)] = -4 - 3(4)(-6)$$

$$2y'' - 6[12 - 72 + 30] = 682y'' + 180 = 68y'' = -56$$

—

## Question 3: Parametric Equations

**Problem:** Find the first and second derivatives of $x = 3\cos\theta - \cos^3\theta$ and $y = 3\sin\theta - \sin^3\theta$.

**Solution:** First, find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$:

$$\frac{dx}{d\theta} = -3\sin\theta + 3\cos^2\theta\sin\theta = -3\sin\theta(1-\cos^2\theta) = -3\sin^3\theta$$

$$\frac{dy}{d\theta} = 3\cos\theta - 3\sin^2\theta\cos\theta = 3\cos\theta(1-\sin^2\theta) = 3\cos^3\theta$$

The first derivative is:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3\cos^3\theta}{-3\sin^3\theta} = -\cot^3\theta$$

The second derivative is:

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\cdot\frac{d\theta}{dx}$$

$$\frac{d}{d\theta}(-\cot^3\theta) = -3\cot^2\theta(-\csc^2\theta) = 3\cot^2\theta\csc^2\theta$$

$$\frac{d^2y}{dx^2} = (3\cot^2\theta\csc^2\theta)\cdot\frac{1}{-3\sin^3\theta} = -\cot^2\theta\csc^5\theta$$

—

## Question 5: Proof

**Problem:** If $\sin y = x\sin(a+y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$.

**Solution:** Isolate $x$:

$$x = \frac{\sin y}{\sin(a+y)}$$

Differentiate $x$ with respect to $y$ using the quotient rule:

$$\frac{dx}{dy} = \frac{(\cos y)\sin(a+y) - (\sin y)\cos(a+y)}{\sin^2(a+y)}$$

Using the identity $\sin(A-B) = \sin A\cos B - \cos A\sin B$:

$$\frac{dx}{dy} = \frac{\sin((a+y)-y)}{\sin^2(a+y)} = \frac{\sin a}{\sin^2(a+y)}$$

Inverting the result to find $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

This completes the proof.

**Question 1**

Geometrically integrating $\int_a^b f(x)\,dx$ means finding the

[label=(A)]

1. area under the curve from a to b
2. area to the left of point a
3. area to the right of point b
4. area above the curve from a to b

**Solution:**

- **Rule Used:** Definition of a Definite Integral.
- **Step 1:** The definite integral of a function $f(x)$ between two limits, $a$ and $b$, calculates the accumulated area bounded by the graph of the function and the x-axis, between the vertical lines $x = a$ and $x = b$.
- **Answer: (A) area under the curve from a to b**

---

**Question 2**

Which of the following is NOT true

[label=(A)]

1. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$ *(Note: The image's print is slightly blurred, but this is the standard true formula)*
2. $\int \frac{1}{x}\,dx = \log_e x + c$
3. $\int x^n\,dx = \frac{x^{n-1}}{n+1}$
4. $\int \cosh x\,dx = \sinh x + c$

**Solution:**

- **Rule Used:** Power Rule of Integration and Properties of Indefinite Integrals.
- **Step 1:** Let's look at the Power Rule for integration, which states that $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$.
- **Step 2:** Option (C) has the wrong exponent (it says $n - 1$ instead of $n+1$) AND it completely forgets the "$+c$" (constant of integration). Every indefinite integral must have a "$+c$".
- **Answer: (C)** is NOT true.

---

**Question 3**

Integrate $\frac{1}{x^3} - 3x^5 + \sqrt{x} - 1$

[label=(A)]

1. $\frac{x^4}{4} + \frac{4x^3}{3} + \frac{6x^2}{2} + 3x + c$
2. $\frac{-1}{2x^2} - \frac{x^6}{2} + \frac{2}{3}x^{\frac{3}{2}} - x + c$
3. $e^x - \cos x - 4\sin x + c$
4. $e^x + \cos x + 4\sin x + c$

**Solution:**

- **Rule Used:** Power Rule ($\int x^n\,dx = \frac{x^{n+1}}{n+1}$), Sum/Difference Rule (integrate each term separately).
- **Step 1:** Rewrite the expression using exponents so it's easier to integrate. $\frac{1}{x^3} = x^{-3}$ and $\sqrt{x} = x^{\frac{1}{2}}$.
  The expression becomes: $x^{-3} - 3x^5 + x^{\frac{1}{2}} - 1$
- **Step 2:** Integrate term by term:
  - $\int x^{-3}\,dx = \frac{x^{-3+1}}{-3+1} = \frac{x^{-2}}{-2} = -\frac{1}{2x^2}$
  - $\int -3x^5\,dx = -3\left(\frac{x^6}{6}\right) = -\frac{x^6}{2}$
  - $\int x^{\frac{1}{2}}\,dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3}x^{\frac{3}{2}}$
  - $\int -1\,dx = -x$
- **Step 3:** Combine them and add the constant $+c$:

$$-\frac{1}{2x^2} - \frac{x^6}{2} + \frac{2}{3}x^{\frac{3}{2}} - x + c$$

- **Answer: (B)**

---

**Question 4**

Integrate $x^3 + 4x^2 + 6x + 3$
*(Note: There is a typo in the original image where it says $x^3 4x^2$, missing the "+" sign. Based on the options, it is intended to be addition).*

[label=(A)]

1. $\frac{x^4}{4} + \frac{4x^3}{3} + \frac{6x^2}{2} + 3x + c$

2. $\frac{-1}{2x^2} - \frac{x^6}{2} + \frac{2}{3x^2} - x + c$
3. $e^x - \cos x - 4\sin x + c$
4. $e^x + \cos x + 4\sin x + c$

**Solution:**

- **Rule Used:** Power Rule of Integration, Constant Rule.
- **Step 1:** Integrate term by term by adding 1 to the power and dividing by the new power.
  - $\int x^3\,dx = \frac{x^4}{4}$
  - $\int 4x^2\,dx = \frac{4x^3}{3}$
  - $\int 6x\,dx = \frac{6x^2}{2}$
  - $\int 3\,dx = 3x$
- **Step 2:** Add them together with $+c$.
- **Answer: (A)**

---

**Question 5**

Integrate $e^x - \sin x + 4\cos x$

[label=(A)]

1. $\frac{x^4}{4} + \frac{4x^3}{3} + \frac{6x^2}{2} + 3x + c$
2. $\frac{-1}{2x^2} - \frac{x^6}{2} + \frac{2}{3x^2} - x + c$
3. $e^x - \cos x - 4\sin x + c$
4. $e^x + \cos x + 4\sin x + c$

**Solution:**

- **Rule Used:** Standard Trigonometric and Exponential Integrals.
- **Step 1:** Know your basic rules:
  - $\int e^x\,dx = e^x$
  - $\int -\sin x\,dx = \cos x$ (because the derivative of $\cos x$ is $-\sin x$)
  - $\int \cos x\,dx = \sin x$, so $\int 4\cos x\,dx = 4\sin x$
- **Step 2:** Combine the results: $e^x + \cos x + 4\sin x + c$.
- **Answer: (D)**

---

**Question 6**

Integrate $\int x^2 \sin x^3 \, dx$

[label=(A)]

1. $\frac{1}{22}(x^2+7)^{11} + c$
2. $-\frac{1}{3}\cos x^3 + c$
3. $-\frac{\cos^2 x}{2} + c$
4. $\frac{\cos^2 x}{2} + c$

**Solution:**

- **Rule Used:** U-Substitution (Reverse Chain Rule).
- **Step 1:** Notice that the derivative of $x^3$ (which is $3x^2$) is very close to the $x^2$ sitting outside the sine function. This means we use u-substitution. Let $u = x^3$.
- **Step 2:** Differentiate $u$: $du = 3x^2\,dx$. To isolate the $x^2\,dx$ that we have in our problem, divide by 3: $\frac{du}{3} = x^2\,dx$.
- **Step 3:** Substitute back into the integral:

$$\int (\sin u) \cdot \frac{du}{3} = \frac{1}{3}\int \sin u \, du$$

- **Step 4:** Integrate: $\frac{1}{3}(-\cos u) + c = -\frac{1}{3}\cos u + c$.
- **Step 5:** Replace $u$ with $x^3$: $-\frac{1}{3}\cos(x^3) + c$.
- **Answer: (B)**

---

**Question 7**

Integrate $\int \sin x \cos x \, dx$

[label=(A)]

1. $\frac{1}{22}(x^2+7)^{11} + c$
2. $-\frac{1}{3}\cos x^3 + c$
3. $-\frac{\cos^2 x}{2} + c$
4. $\frac{\cos^2 x}{2} + c$

**Solution:**

- **Rule Used:** U-Substitution.
- **Step 1:** Let $u = \cos x$.
- **Step 2:** Find the derivative: $du = -\sin x\,dx$. Therefore, $-du = \sin x\,dx$.
- **Step 3:** Substitute into the integral: $\int u(-du) = -\int u\,du$.
- **Step 4:** Apply power rule: $-\frac{u^2}{2} + c$.
- **Step 5:** Replace $u$ with $\cos x$: $-\frac{\cos^2 x}{2} + c$.
- **Answer: (C)**

---

**Question 8 & 10**

*(Note: Questions 8 and 10 are identical on the paper)*
Integrate $\int \cos^2 x\,dx$

[label=(A)]

1. $\frac{1}{2}x + \frac{1}{4}\sin 2x + c$
2. $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c$
3. $-\frac{\cos^2 x}{2} + c$
4. $\frac{1}{8}x - \frac{1}{32}\sin 4x + c$

**Solution:**

- **Rule Used:** Half-Angle Trigonometric Identity.
- **Step 1:** You cannot integrate $\cos^2 x$ directly using the power rule. Use the identity: $\cos^2 x = \frac{1+\cos 2x}{2}$.
- **Step 2:** Rewrite the integral: $\int \left(\frac{1}{2} + \frac{1}{2}\cos 2x\right)\,dx$.
- **Step 3:** Integrate each part:
  - $\int \frac{1}{2}\,dx = \frac{1}{2}x$
  - $\int \frac{1}{2}\cos 2x\,dx$. Remember $\int \cos(ax)\,dx = \frac{\sin(ax)}{a}$. So, $\frac{1}{2} \cdot \frac{\sin 2x}{2} = \frac{1}{4}\sin 2x$.
- **Step 4:** Combine: $\frac{1}{2}x + \frac{1}{4}\sin 2x + c$.
- **Answer: (A)**

---

**Question 9**

Integrate $\int \sin^4 x \, dx$

[label=(A)]

1. $\frac{1}{2}x + \frac{1}{4}\sin 2x + c$
2. $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c$
3. $-\frac{\cos^2 x}{2} + c$
4. $\frac{1}{8}x - \frac{1}{32}\sin 4x + c$

**Solution:**

- **Rule Used:** Half-Angle Trigonometric Identity (applied twice).
- **Step 1:** Rewrite as $(\sin^2 x)^2$. Using the identity $\sin^2 x = \frac{1-\cos 2x}{2}$, we get:

$$\left(\frac{1-\cos 2x}{2}\right)^2 = \frac{1 - 2\cos 2x + \cos^2 2x}{4}$$

- **Step 2:** Use the identity again for $\cos^2 2x = \frac{1+\cos 4x}{2}$. Substitute this back in:

$$\frac{1}{4}\left(1 - 2\cos 2x + \frac{1+\cos 4x}{2}\right) = \frac{1}{4}\left(\frac{3}{2} - 2\cos 2x + \frac{1}{2}\cos 4x\right)$$

$$= \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$$

- **Step 3:** Integrate term by term:
  - $\int \frac{3}{8}\, dx = \frac{3}{8}x$
  - $\int -\frac{1}{2}\cos 2x \, dx = -\frac{1}{2}\left(\frac{\sin 2x}{2}\right) = -\frac{1}{4}\sin 2x$
  - $\int \frac{1}{8}\cos 4x \, dx = \frac{1}{8}\left(\frac{\sin 4x}{4}\right) = \frac{1}{32}\sin 4x$
- **Step 4:** Combine: $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c$.
- **Answer: (B)**

---

**Question 11**

Evaluate $\int \frac{1}{\sqrt{x}}\,dx$

[label=(A)]

1. $2x + c$
2. $x^2 + c$
3. $2\sqrt{x} + c$
4. $\frac{1}{x} + c$

**Solution:**

- **Rule Used:** Power Rule.
- **Step 1:** Rewrite the root as a fractional exponent and move it to the top: $\frac{1}{x^{0.5}} = x^{-\frac{1}{2}}$.
- **Step 2:** Apply the power rule (add 1 to power, divide by new power):

$$\frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{x^{\frac{1}{2}}}{\frac{1}{2}}$$

- **Step 3:** Dividing by $\frac{1}{2}$ is the same as multiplying by 2. So we get $2x^{\frac{1}{2}} + c$, which is $2\sqrt{x} + c$.
- **Answer: (C)**

---

**Question 12**

Determine $\int_0^1 x(x^2 + 1)\,dx$

[label=(A)]

1. $\frac{1}{4}$
2. $\frac{7}{2}$
3. $\frac{3}{4}$
4. $-\frac{2}{5}$

**Solution:**

- **Rule Used:** Algebraic Expansion and Fundamental Theorem of Calculus.
- **Step 1:** Expand the expression first to avoid complicated rules: $x(x^2) + x(1) = x^3 + x$.

- **Step 2:** Integrate using the Power Rule: $\int (x^3 + x)\,dx = \left[\frac{x^4}{4} + \frac{x^2}{2}\right]_0^1$
- **Step 3:** Evaluate at the upper limit (1) and subtract the lower limit (0):

$$\left(\frac{1^4}{4} + \frac{1^2}{2}\right) - \left(\frac{0^4}{4} + \frac{0^2}{2}\right) = \frac{1}{4} + \frac{1}{2} = \frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

- **Answer: (C)**

---

**Question 13**

Obtain $\int_{-1}^{2}(x^2 - 2x + 3)\,dx$

[label=(A)]

1. 2
2. $-1$
3. 1
4. 9

**Solution:**

- **Rule Used:** Power Rule and Fundamental Theorem of Calculus.
- **Step 1:** Integrate the polynomial term by term: $\left[\frac{x^3}{3} - x^2 + 3x\right]_{-1}^{2}$
- **Step 2:** Evaluate at the upper limit (2):

$$\left(\frac{2^3}{3} - 2^2 + 3(2)\right) = \left(\frac{8}{3} - 4 + 6\right) = \frac{8}{3} + 2 = \frac{8}{3} + \frac{6}{3} = \frac{14}{3}$$

- **Step 3:** Evaluate at the lower limit (-1):

$$\left(\frac{(-1)^3}{3} - (-1)^2 + 3(-1)\right) = \left(-\frac{1}{3} - 1 - 3\right) = -\frac{1}{3} - 4 = -\frac{1}{3} - \frac{12}{3} = -\frac{13}{3}$$

- **Step 4:** Subtract the lower evaluation from the upper:

$$\frac{14}{3} - \left(-\frac{13}{3}\right) = \frac{14}{3} + \frac{13}{3} = \frac{27}{3} = 9$$

- **Answer: (D)**

---

**Question 14**

Evaluate $\int_0^1 (3x+1)^5\,dx$ to four significant figures

[label=(A)]

1. 277.7
2. 270.5
3. 277.5
4. 227.6

**Solution:**

- **Rule Used:** U-Substitution.
- **Step 1:** Let $u = 3x + 1$. Then $du = 3\,dx$, meaning $dx = \frac{du}{3}$.
- **Step 2:** Change the limits. When $x = 0$, $u = 3(0) + 1 = 1$. When $x = 1$, $u = 3(1) + 1 = 4$.
- **Step 3:** Substitute into the integral: $\int_1^4 u^5 \frac{du}{3} = \frac{1}{3}\int_1^4 u^5\,du$.
- **Step 4:** Integrate and evaluate:

$$\frac{1}{3}\left[\frac{u^6}{6}\right]_1^4 = \frac{1}{18}[u^6]_1^4 = \frac{1}{18}(4^6 - 1^6)$$

$$= \frac{1}{18}(4096 - 1) = \frac{4095}{18} = 227.5$$

- *Note: The exact answer is 227.5. Looking at the options, there might be a typo in the provided options as (C) is 277.5 and (D) is 227.6. The closest mathematical evaluation is exactly 227.5.*

---

**Question 15**

Evaluate $\int 3\sqrt{x}\,dx$

[label=(A)]

1. $\sqrt{x^2} + c$
2. $2\sqrt{x^3} + c$
3. $2x\sqrt{x^3} + c$
4. $2x^2\sqrt{x^3} + c$

**Solution:**

- **Rule Used:** Power Rule.
- **Step 1:** Rewrite as an exponent: $\int 3x^{\frac{1}{2}}\,dx$.
- **Step 2:** Apply the power rule:

$$3\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)+c=3\cdot\frac{2}{3}x^{\frac{3}{2}}+c=2x^{\frac{3}{2}}+c$$

- **Step 3:** Convert back to a radical format. $x^{\frac{3}{2}}$ is the same as $\sqrt{x^3}$. Result: $2\sqrt{x^3}+c$.
- **Answer: (B)**

---

**Question 16**

Find $\int\left(\frac{2m^2+1}{m}\right)\,dm$

[label=(A)]

1. $m^3+\ln m+c$
2. $m^2+m\ln m+c$
3. $m^3+\ln m+c$
4. $m^2+\ln m+c$

**Solution:**

- **Rule Used:** Algebraic simplification and Natural Log Rule ($\int\frac{1}{x}\,dx=\ln|x|$).
- **Step 1:** Split the fraction to simplify: $\frac{2m^2}{m}+\frac{1}{m}=2m+\frac{1}{m}$.
- **Step 2:** Integrate term by term:
  - $\int 2m\,dm=2\left(\frac{m^2}{2}\right)=m^2$
  - $\int\frac{1}{m}\,dm=\ln(m)$
- **Step 3:** Combine with $+c$: $m^2+\ln m+c$.
- **Answer: (D)**

---

**Question 17**

Determine $\int \sqrt{3-4x}\,dx$
*(No options were visible for this question, but here is the solution).*

**Solution:**

- **Rule Used:** U-Substitution.
- **Step 1:** Rewrite as $\int (3-4x)^{\frac{1}{2}}\,dx$. Let $u = 3-4x$.
- **Step 2:** Find derivative: $du = -4\,dx$, so $dx = -\frac{du}{4}$.
- **Step 3:** Substitute: $\int u^{\frac{1}{2}}\left(-\frac{du}{4}\right) = -\frac{1}{4}\int u^{\frac{1}{2}}\,du$.
- **Step 4:** Apply power rule:

$$-\frac{1}{4}\left(\frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right) + c = -\frac{1}{4}\cdot\frac{2}{3}u^{\frac{3}{2}} + c = -\frac{1}{6}u^{\frac{3}{2}} + c$$

- **Step 5:** Substitute back $u = 3-4x$:

$$-\frac{1}{6}(3-4x)^{\frac{3}{2}} + c$$

## Question 18

**Evaluate** $\int_{-2}^{3}(4-x^2)dx$

**Options:**

[label=**A.**]

1. $8\frac{1}{3}$
2. $2\frac{1}{6}$
3. $11\frac{1}{4}$
4. $5\frac{1}{3}$
5. $\frac{1}{3}$

**Solution:**

- **Rule Used:** Power Rule for Integration ($\int x^n dx = \frac{x^{n+1}}{n+1}$) and the Fundamental Theorem of Calculus (plugging in limits).
- **Step 1:** Find the integral of each term separately. The integral of 4 is $4x$, and the integral of $x^2$ is $\frac{x^3}{3}$.

$$\int (4-x^2)dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{3}$$

- **Step 2:** Plug in the upper limit ($x = 3$).

$$4(3) - \frac{3^3}{3} = 12 - \frac{27}{3} = 12 - 9 = 3$$

- **Step 3:** Plug in the lower limit ($x = -2$).

$$4(-2) - \frac{(-2)^3}{3} = -8 - \frac{-8}{3} = -8 + \frac{8}{3} = \frac{-24}{3} + \frac{8}{3} = -\frac{16}{3}$$

- **Step 4:** Subtract the lower limit result from the upper limit result.

$$3 - \left(-\frac{16}{3}\right) = 3 + \frac{16}{3} = \frac{9}{3} + \frac{16}{3} = \frac{25}{3} = 8\frac{1}{3}$$

**Answer: A**

---

## Question 19

**Find** $\int_1^3 \left(\frac{2x^2+1}{x}\right) dx$

**Options:**

[label=**A.**]

1. 0.1926
2. 98.25
3. 9.099
4. 59.64
5. 86.25

**Solution:**

- **Rule Used:** Algebraic Simplification, Power Rule, and Logarithmic Rule for Integration ($\int \frac{1}{x} dx = \ln|x|$).
- **Step 1:** Separate the fraction into two simpler terms.

$$\frac{2x^2}{x} + \frac{1}{x} = 2x + \frac{1}{x}$$

- **Step 2:** Integrate the simplified expression.

$$\int \left(2x + \frac{1}{x}\right) dx = \left[x^2 + \ln|x|\right]_1^3$$

- **Step 3:** Evaluate using the limits (3 and 1).

$$Upper : (3^2 + \ln 3) = 9 + 1.0986 = 10.0986$$

$$Lower : (1^2 + \ln 1) = 1 + 0 = 1$$

- **Step 4:** Subtract.

$$10.0986 - 1 = 9.0986 \approx 9.099$$

**Answer: C**

---

## Question 20

**Evaluate** $\int_0^1 2e^{6x-1}dx$

**Options:**

[label=**A.**]

1. 49.35
2. 57.37
3. 239.3
4. 409.4
5. 149.4

**Solution:**

- **Rule Used:** Exponential Rule for Integration ($\int e^{kx}dx = \frac{1}{k}e^{kx}$).
- **Step 1:** Integrate the function. The derivative of the exponent $(6x - 1)$ is 6, so we divide by 6.

$$\int 2e^{6x-1}dx = \frac{2}{6}e^{6x-1} = \left[\frac{1}{3}e^{6x-1}\right]_0^1$$

- **Step 2:** Plug in the limits.

$$Upper(x = 1) : \frac{1}{3}e^{6(1)-1} = \frac{1}{3}e^5 \approx \frac{148.413}{3} \approx 49.471$$

$$Lower(x = 0) : \frac{1}{3}e^{6(0)-1} = \frac{1}{3}e^{-1} \approx \frac{0.3678}{3} \approx 0.1226$$

- **Step 3:** Subtract.

$$49.471 - 0.1226 = 49.3484 \approx 49.35$$

**Answer: A**

---

## Question 21

**Determine** $\int_0^1 \frac{3x}{(4x^2-1)^5}dx$

**Options:**

[label=**A.**]

1. 0.9559
2. 0.09259
3. 0.02599
4. 2581
5. 1089

**Solution:**

- **Rule Used:** Integration by Substitution (U-substitution). *(Note: The denominator equals zero at $x = 0.5$, which makes this an improper integral mathematically, but standard testing usually expects you to evaluate it blindly using algebraic substitution.)*
- **Step 1:** Let $u = 4x^2 - 1$. Then, the derivative $du = 8x\,dx$, which means $x\,dx = \frac{du}{8}$.
- **Step 2:** Change the bounds. If $x = 0$, $u = -1$. If $x = 1$, $u = 3$.
- **Step 3:** Substitute into the integral.

$$\int_{-1}^{3} \frac{3}{u^5}\left(\frac{du}{8}\right) = \frac{3}{8}\int_{-1}^{3} u^{-5}du$$

- **Step 4:** Integrate using the Power Rule.

$$\frac{3}{8}\left[\frac{u^{-4}}{-4}\right]_{-1}^{3} = -\frac{3}{32}\left[\frac{1}{u^4}\right]_{-1}^{3}$$

- **Step 5:** Evaluate the bounds.

$$-\frac{3}{32}\left(\frac{1}{3^4} - \frac{1}{(-1)^4}\right) = -\frac{3}{32}\left(\frac{1}{81} - 1\right) = -\frac{3}{32}\left(-\frac{80}{81}\right)$$

- **Step 6:** Simplify.

$$\frac{3 \times 80}{32 \times 81} = \frac{240}{2592} = \frac{10}{108} = \frac{5}{54} \approx 0.09259$$

**Answer: B**

---

## Question 22

**Find** $\int \frac{2e^t}{\sqrt{e^t+4}}dt$

**Options:**

[label=**A.**]

1. $6\sqrt{e^t+2}+c$
2. $\sqrt{e^t+4}+c$
3. $4\sqrt{e^t+4}+c$
4. $4\sqrt{e^{2t}-4}+c$
5. $2\sqrt{e^t+16}+c$

**Solution:**

- **Rule Used:** U-substitution.
- **Step 1:** Let the inside of the root be $u$. So, $u = e^t + 4$. The derivative is $du = e^t\,dt$.
- **Step 2:** Substitute into the integral. The $e^t\,dt$ becomes $du$.

$$\int \frac{2}{\sqrt{u}}du = \int 2u^{-1/2}du$$

- **Step 3:** Integrate using the Power Rule (add 1 to power, divide by new power).

$$2\left(\frac{u^{1/2}}{1/2}\right) + c = 4\sqrt{u} + c$$

- **Step 4:** Replace $u$ with $e^t + 4$.

$$4\sqrt{e^t+4}+c$$

**Answer: C**

---

## Question 23

**Determine** $\int 3x(4x^2+3)^5dx$

**Options:**

[label=**A.**]

1. $\frac{(12x^2+9)^6}{6}+c$
2. $\frac{(4x^2+3)^6}{48}+c$

3. $\frac{(4x^2+3)^4}{64} + c$
4. $\frac{(4x^2+3)^6}{16} + c$
5. $(4x^2+3)^6 + c$

**Solution:**

- **Rule Used:** U-substitution.
- **Step 1:** Let the inside of the parenthesis be $u$. $u = 4x^2+3$. The derivative is $du = 8x\,dx$, so $x\,dx = \frac{du}{8}$.
- **Step 2:** Substitute everything.

$$\int 3\left(\frac{du}{8}\right)u^5 = \frac{3}{8}\int u^5 du$$

- **Step 3:** Integrate using the Power Rule.

$$\frac{3}{8}\left(\frac{u^6}{6}\right) + c = \frac{3}{48}u^6 + c = \frac{1}{16}u^6 + c$$

- **Step 4:** Replace $u$.

$$\frac{(4x^2+3)^6}{16} + c$$

**Answer: D**

---

## Question 24

**Determine** $\int \frac{4}{5x-3}dx$

**Options:**

[label=**A.**]

1. $(5x-3) + c$
2. $\ln(5x-3)$
3. $\frac{4}{5}\ln(5x-3) + c$
4. $\frac{(5x-3)}{5} + c$
5. $\frac{15}{4}\ln(5x-3) + c$

**Solution:**

- **Rule Used:** Logarithmic Rule for Integration ($\int \frac{1}{ax+b}dx = \frac{1}{a}\ln|ax+b|$).

- **Step 1:** Pull the constant 4 out of the integral.

$$4\int \frac{1}{5x-3}dx$$

- **Step 2:** Apply the rule directly. Since the coefficient of $x$ is 5, we divide by 5 and take the natural log of the denominator.

$$4\left(\frac{1}{5}\ln|5x-3|\right)+c=\frac{4}{5}\ln(5x-3)+c$$

**Answer: C**

---

## Question 25

**Obtain** $\int \frac{2x}{\sqrt{4x^2-1}}dx$

**Options:**

[label=**A.**]

1. $x(4x^2-1)+c$
2. $\frac{\sqrt{4x^2-1}}{2}+c$
3. $\frac{(4x^2-1)}{8}+c$
4. $\sqrt{4x^2-1}+c$
5. $\frac{\sqrt{x^2-1}}{2}+c$

**Solution:**

- **Rule Used:** U-substitution.
- **Step 1:** Let $u=4x^2-1$. Then $du=8x\,dx$. We have $2x\,dx$ in the numerator, so we divide by 4: $\frac{du}{4}=2x\,dx$.
- **Step 2:** Substitute into the integral.

$$\int \frac{1}{\sqrt{u}}\left(\frac{du}{4}\right)=\frac{1}{4}\int u^{-1/2}du$$

- **Step 3:** Integrate.

$$\frac{1}{4}\left(\frac{u^{1/2}}{1/2}\right)+c=\frac{1}{4}\left(2\sqrt{u}\right)+c=\frac{1}{2}\sqrt{u}+c$$

- **Step 4:** Replace $u$.

$$\frac{\sqrt{4x^2-1}}{2}+c$$

**Answer: B**

---

## Question 26

**Determine** $\int \tan\theta d\theta$

**Options:**

[label=**A.**]

1. $-\ln\sec\theta + c$
2. $\sec^2\theta + c$
3. $\ln\sec\theta + c$
4. $\ln\csc\theta + c$
5. $\cot\theta + c$

**Solution:**

- **Rule Used:** Trigonometric Identity and U-substitution.
- **Step 1:** Rewrite tangent as sine over cosine: $\int \frac{\sin\theta}{\cos\theta} d\theta$.
- **Step 2:** Let $u = \cos\theta$. Then $du = -\sin\theta\, d\theta$, which means $-du = \sin\theta\, d\theta$.
- **Step 3:** Substitute.

$$\int -\frac{1}{u} du = -\ln|u| + c = -\ln|\cos\theta| + c$$

- **Step 4:** Using logarithm rules, $-\ln|\cos\theta| = \ln|(\cos\theta)^{-1}| = \ln|\sec\theta| + c$.

**Answer: C**

---

## Question 27

**Simplify** $\int_{\pi/2}^{\pi} 3\sin 2x dx$

**Options:**

[label=**A.**]

1. 3
2. 4
3. 9
4. 6
5. 4

**Solution:**

- **Rule Used:** Integration of Trigonometric Functions ($\int \sin(kx)dx = -\frac{1}{k}\cos(kx)$).
- **Step 1:** Integrate the function.

$$\int 3\sin 2x dx = 3\left(-\frac{1}{2}\cos 2x\right) = -\frac{3}{2}\cos 2x$$

- **Step 2:** Evaluate from $\pi/2$ to $\pi$.

$$Upper(x=\pi): -\frac{3}{2}\cos(2\pi) = -\frac{3}{2}(1) = -1.5$$

$$Lower(x=\pi/2): -\frac{3}{2}\cos\left(2\left(\frac{\pi}{2}\right)\right) = -\frac{3}{2}\cos(\pi) = -\frac{3}{2}(-1) = 1.5$$

- **Step 3:** Subtract the lower limit from upper limit.

$$-1.5 - (1.5) = -3$$

*(Note: It appears there is a typo in the provided test paper limits. If the lower limit were* 0 *instead of* $\pi/2$*, the answer would perfectly equal* 3*. We will select the corresponding matching magnitude from the option set).*

**Likely Intended Answer: A** (assuming limits were intended as 0 to $\pi/2$).

---

## Question 28

**The area bounded by the curve** $y = 8 + 2x - x^2$ **between** $x = -2$ **and** $x = 4$ **is?**

**Options:**

[label=**A.**]

1. 30 Sq. Units
2. 56 Sq. Units
3. 36 Sq. Units
4. 70 Sq. Units
5. 15 Sq. Units

**Solution:**

- **Rule Used:** Definite Integral for Area under a curve ($Area = \int_a^b f(x)dx$).
- **Step 1:** Set up the integral.

$$\int_{-2}^{4}(8 + 2x - x^2)dx$$

- **Step 2:** Integrate term by term.

$$\left[8x + x^2 - \frac{x^3}{3}\right]_{-2}^{4}$$

- **Step 3:** Evaluate at upper limit (4).

$$8(4) + (4)^2 - \frac{4^3}{3} = 32 + 16 - \frac{64}{3} = 48 - 21.33 = 26.67$$

- **Step 4:** Evaluate at lower limit $(-2)$.

$$8(-2) + (-2)^2 - \frac{(-2)^3}{3} = -16 + 4 - \left(-\frac{8}{3}\right) = -12 + 2.67 = -9.33$$

- **Step 5:** Subtract.

$$26.67 - (-9.33) = 26.67 + 9.33 = 36$$

**Answer: C**

---

## Question 29

**Find** $\int_{-1}^{2} \frac{2}{3e^{2x}} dx$

**Options:**

[label=**A.**]

1. 20.5
2. 2.45
3. 78.55
4. 7.27
5. 90.2

**Solution:**

- **Rule Used:** Negative Exponent Rule and Exponential Integration.
- **Step 1:** Rewrite the fraction to move the exponent to the numerator.

$$\int_{-1}^{2} \frac{2}{3} e^{-2x} dx$$

- **Step 2:** Integrate. Divide by the derivative of the exponent $(-2)$.

$$\left[\frac{2}{3}\left(\frac{e^{-2x}}{-2}\right)\right]_{-1}^{2} = \left[-\frac{1}{3}e^{-2x}\right]_{-1}^{2}$$

- **Step 3:** Evaluate the bounds.

$$Upper(2) : -\frac{1}{3}e^{-4} \approx -0.0061$$

$$Lower(-1) : -\frac{1}{3}e^{2} \approx -\frac{7.389}{3} \approx -2.463$$

- **Step 4:** Subtract.

$$-0.0061 - (-2.463) = 2.4569 \approx 2.45$$

**Answer: B**

---

## Question 30

**A public toilet is treated periodically to control the growth of harmful bacteria. The concentration of bacteria per cm$^3$ after $t$ days is given by $C(t) = 30t^2 - 300t + 610$. In how many days after a treatment will the concentration be minimal?**

**Options:**

[label=**A.**]

1. 4 days
2. 2 days
3. 5 days
4. 7 days
5. 15 days

**Solution:**

- **Rule Used:** Optimization using the First Derivative (a function hits a minimum/maximum when its first derivative equals 0).
- **Step 1:** Find the derivative of $C(t)$ with respect to $t$.

$$C'(t) = 60t - 300$$

- **Step 2:** Set the derivative equal to 0 and solve for $t$.

$$60t - 300 = 0$$

$$60t = 300$$

$$t = \frac{300}{60} = 5$$

**Answer: C**

---

## Question 31

**Simplify** $\int_1^3 \frac{2m^2+1}{m} dm$

**Options:**

[label=**A.**]

1. 900.1
2. 78.65
3. 9.099
4. 6.089
5. 5.813

**Solution:** *(Note: This is the exact same mathematical problem as Question 19, just using the variable 'm' instead of 'x'.)*

- **Step 1:** Split fraction: $2m + \frac{1}{m}$
- **Step 2:** Integrate: $[m^2 + \ln m]_1^3$
- **Step 3:** Evaluate: $(3^2 + \ln 3) - (1^2 + \ln 1) = (9 + 1.0986) - 1 = 9.0986 \approx 9.099$.

**Answer: C**

---

## Question 32

**At what rate is the area of a circle changing when the radius is 4cm and changes at the rate of 0.5 cm/s?**

**Options:**

[label=**A.**]

1. $4\pi cm^2/sec$
2. $7\pi cm^2/sec$
3. $9\pi cm^2/sec$
4. $8\pi cm^2/sec$
5. $10\pi cm^2/sec$

**Solution:**

- **Rule Used:** Related Rates using the Chain Rule.
- **Step 1:** Write the formula for the Area of a circle.

$$A = \pi r^2$$

- **Step 2:** Differentiate both sides with respect to time ($t$).

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt}$$

- **Step 3:** Substitute the given values: radius $r = 4$, rate of radius change $\frac{dr}{dt} = 0.5$.

$$\frac{dA}{dt} = 2\pi(4)(0.5) = 8\pi(0.5) = 4\pi$$

**Answer: A**

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## Question 33

**Evaluate** $\int_0^{\pi/2} \cos\theta d\theta$

**Options:**

[label=**(A)**]

1. 0
2. 3
3. 0.5
4. 1.0

**Solution:**

- **Rule Used:** Basic Trigonometric Integral ($\int \cos\theta d\theta = \sin\theta$).
- **Step 1:** Integrate.

$$[\sin\theta]_0^{\pi/2}$$

- **Step 2:** Evaluate the bounds.

$$\sin\left(\frac{\pi}{2}\right) - \sin(0)$$

$$1 - 0 = 1$$

**Answer: D**

---

## Question 34

**Determine $\int_0^1 2e^{6x-1}dx$, correct to 4 significant figures**

**Options:**

[label=**(A)**]

1. 49.35
2. 37.45
3. 21.97
4. 31.65

**Solution:** *(Note: This is the same problem as Question 20).*

- **Calculation:**
$$\left[\frac{1}{3}e^{6x-1}\right]_0^1 = \frac{1}{3}e^5 - \frac{1}{3}e^{-1} \approx 49.3484$$
- Rounded to 4 significant figures, it is 49.35.

**Answer: A**

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## Question 35

**Evaluate $\int_1^2 (3x^4 - x^5)dx$**

**Options:**

[label=**(A)**]

1. $\frac{81}{10}$
2. $6\frac{1}{2}$
3. $\frac{5}{62}$
4. 81

**Solution:**

- **Rule Used:** Power Rule for Integration.
- **Step 1:** Integrate the terms.
$$\left[\frac{3x^5}{5} - \frac{x^6}{6}\right]_1^2$$
- **Step 2:** Evaluate at upper limit (2).
$$\frac{3(2^5)}{5} - \frac{2^6}{6} = \frac{3(32)}{5} - \frac{64}{6} = \frac{96}{5} - \frac{32}{3} = \frac{288}{15} - \frac{160}{15} = \frac{128}{15}$$

- **Step 3:** Evaluate at lower limit (1).

$$\frac{3(1^5)}{5} - \frac{1^6}{6} = \frac{3}{5} - \frac{1}{6} = \frac{18}{30} - \frac{5}{30} = \frac{13}{30}$$

- **Step 4:** Subtract lower from upper limit.

$$\frac{128}{15} - \frac{13}{30} = \frac{256}{30} - \frac{13}{30} = \frac{243}{30}$$

- **Step 5:** Reduce the fraction (divide top and bottom by 3).

$$\frac{81}{10}$$

**Answer: A**